

Users > moshika > Desktop > lab >  hanoi.pl

```
1  % hanoi(N, From, To, Aux)
2  % N    - number of disks
3  % From - source peg
4  % To   - destination peg
5  % Aux  - auxiliary peg
6
7  hanoi(0, _, _, _) :- !.      % Base case: no moves needed
8
9  hanoi(N, From, To, Aux) :-
10     N > 0,
11     N1 is N - 1,
12
13     % Step 1: Move N-1 disks from Source to Aux peg
14     hanoi(N1, From, Aux, To),
15
16     % Step 2: Move the largest disk to Destination peg
17     write('Move disk from '), write(From), write(' to '), write(To), nl,
18
19     % Step 3: Move N-1 disks from Aux to Destination peg
20     hanoi(N1, Aux, To, From).
21
```

```
[moshika@Moshikas-MacBook-Air ~ % cd /Users/moshika/Desktop/lab
[moshika@Moshikas-MacBook-Air lab % swipl hanoi.pl
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.
```

```
For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
```

```
[?- hanoi(3, left, right, middle).
Move disk from left to right
Move disk from left to middle
Move disk from right to middle
Move disk from left to right
Move disk from middle to left
Move disk from middle to right
Move disk from left to right
true.
```

```
[?- halt.
```